

Join the lecture online on *your* dashboard.

Let's start with a minute of silence.

आचार्यात् पादं आधत्ते पादं शिष्यः स्वमेधया । पादं सब्रह्मचारिभ्यः पादं कालक्रमेण च ॥

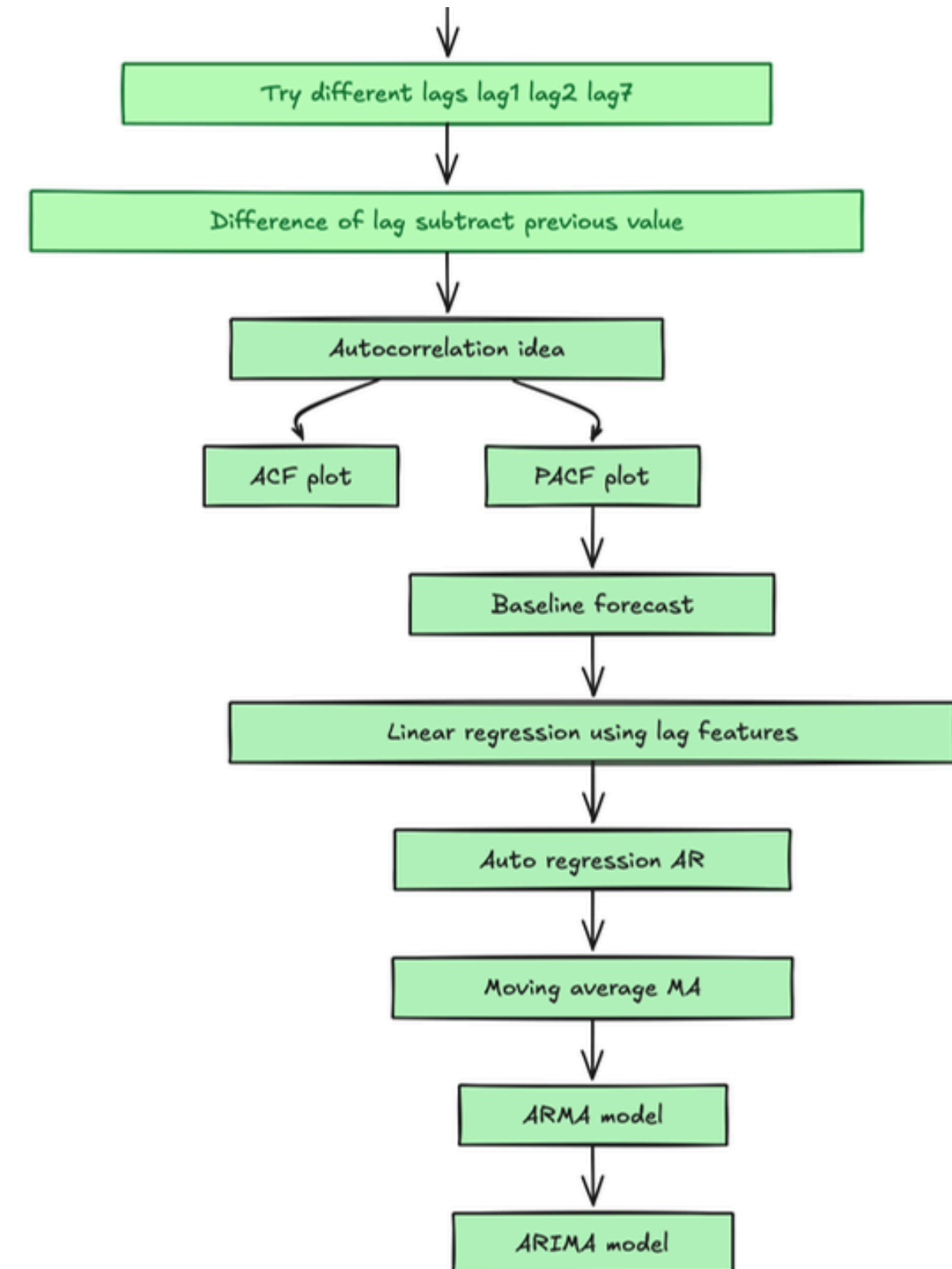
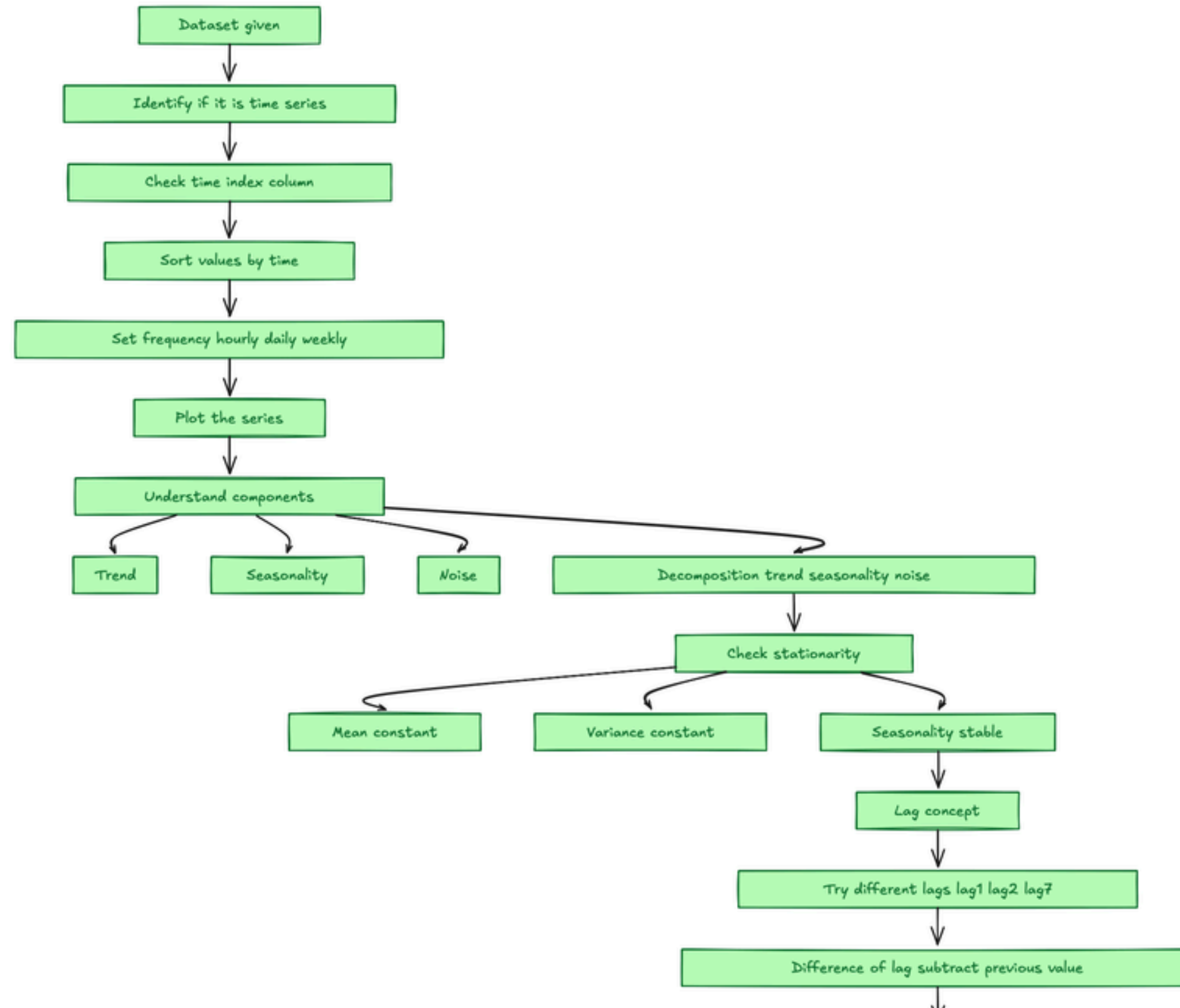
Meaning:

A student acquires knowledge in four equal parts:

- One-fourth from the teacher
- One-fourth through self-reflection and independent thinking
- One-fourth through discussions with peers and fellow learners
- One-fourth over time through personal experience



Recap



Agenda

- **ANNOVA**
- **Problems on ARIMA**
- **SARIMA**
- **Case Study**

Where do p, d, and q come from?

- **d (differencing order)** is chosen by repeatedly differencing the series until the mean becomes stable

→ checked using the time plot, rolling mean/std, and ADF test

- **p (AR order)** is chosen from the PACF plot

→ count the number of significant spikes before PACF cuts off

- **q (MA order)** is chosen from the ACF plot

→ count the number of significant spikes before ACF cuts off



AR(p): depends on past values ⇒ use PACF (direct value impact)

MA(q): depends on past shocks/errors ⇒ use ACF (shock memory)

Which Dhaba will you prefer?



1449790

Which Dhaba will you prefer?



Which Restuarant will you prefer

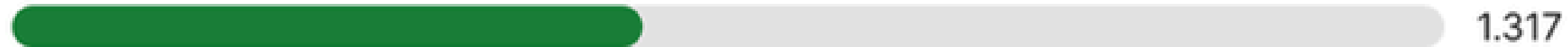
Ranking Poll  63 votes  63 participants

 Share 

1. Neelkanth Star



2. Amrik Sukhdev



3. Mannat Haveli



Problem

Are these restaurants **really different?**

Or are these **small differences** just because **people's mood** changes daily?

Cafe	Average Rating
Neelkanth Star	7.1
Amrik Sukhdev	7.4
Mannat Haveli	7.2

Problem

Are these restaurants **really different**?

- At first glance, Amrik Sukhdev looks best because it has the **highest average**.
- But think carefully:
 - Your mood changes every day
 - Sometimes paratha tastes better, sometimes worse
 - The difference between 7.1 and 7.4 is minimal.
- That difference might exist even if all dabha's are equally good.

Cafe	Average Rating
Neelkanth Star	7.1
Amrik Sukhdev	7.4
Mannat Haveli	7.2

“Is this difference larger than what random variation normally produces?”

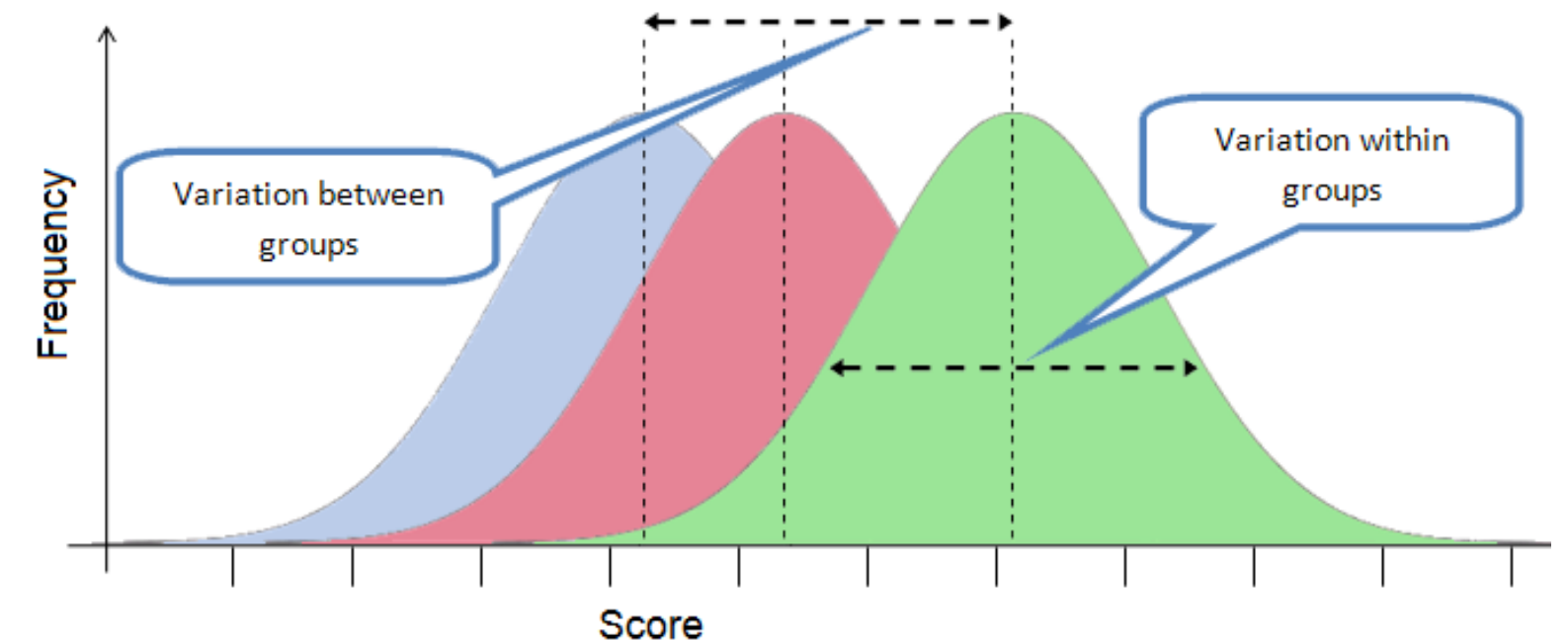
Analysis of Variance(ANOVA)

- ANOVA gives a scientific answer to:
 - Are the differences between **groups** larger than **normal random variation**?
- Use ANOVA when:
 - You **compare 3 or more groups**
 - You **measure numbers (ratings, marks, sales)**
 - You want one **decision** instead of many **guesses**.

Cafe	Average Rating
Neelkanth Star	7.1
Amrik Sukhdev	7.4
Mannat Haveli	7.2

ANOVA

- ANOVA compares two types of variation:
 - **Within-group variation**
 - How much ratings change inside the same Dhabha
 - **Between-group variation**
 - How much Dhabha averages differ from each other
- If **Dhabhas** are truly different:
 - Between-group variation will be much larger



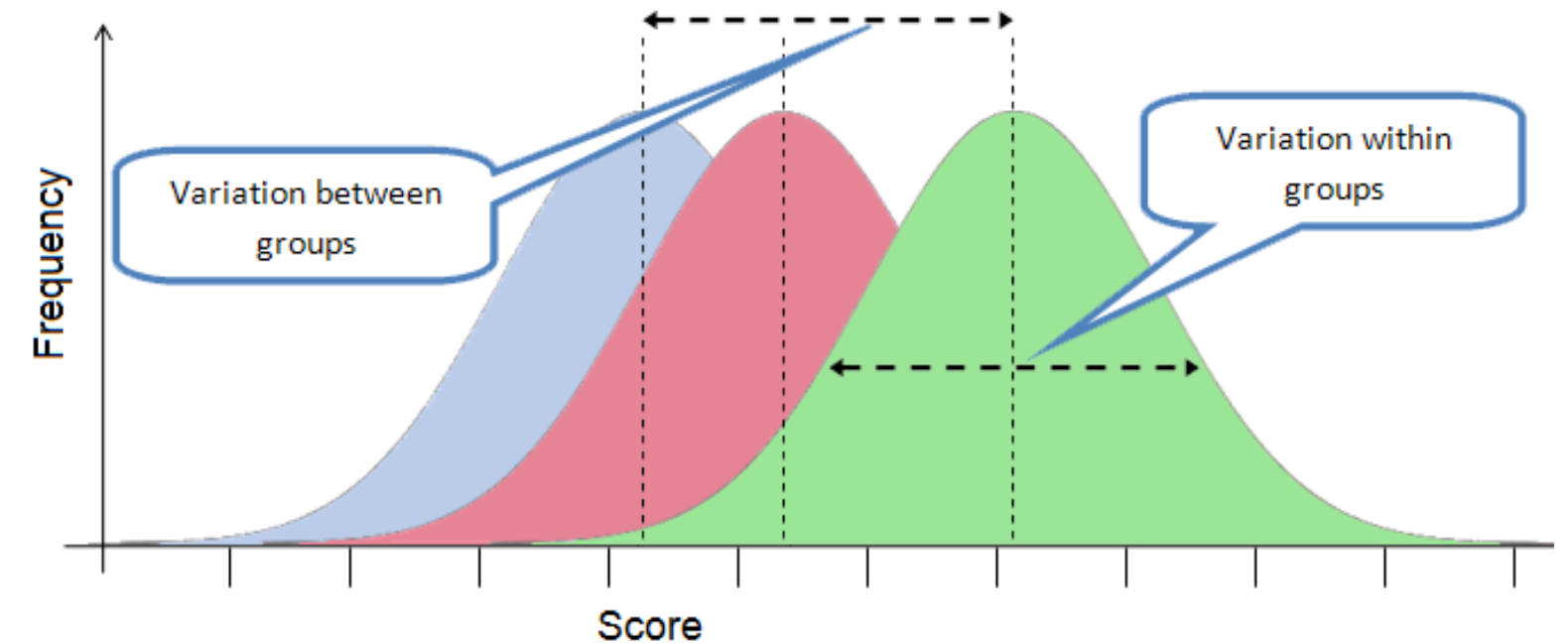
What is “variation”?

- How much do values spread out from an average?
- We measure variation using the sum of squares (SS).

$$SS_{Total} = \sum_{i=1}^N (x_i - \bar{x})^2$$

Where:

- x_i = each observation
- $\bar{x}$ = grand mean (mean of all observations)
- N = total number of observations



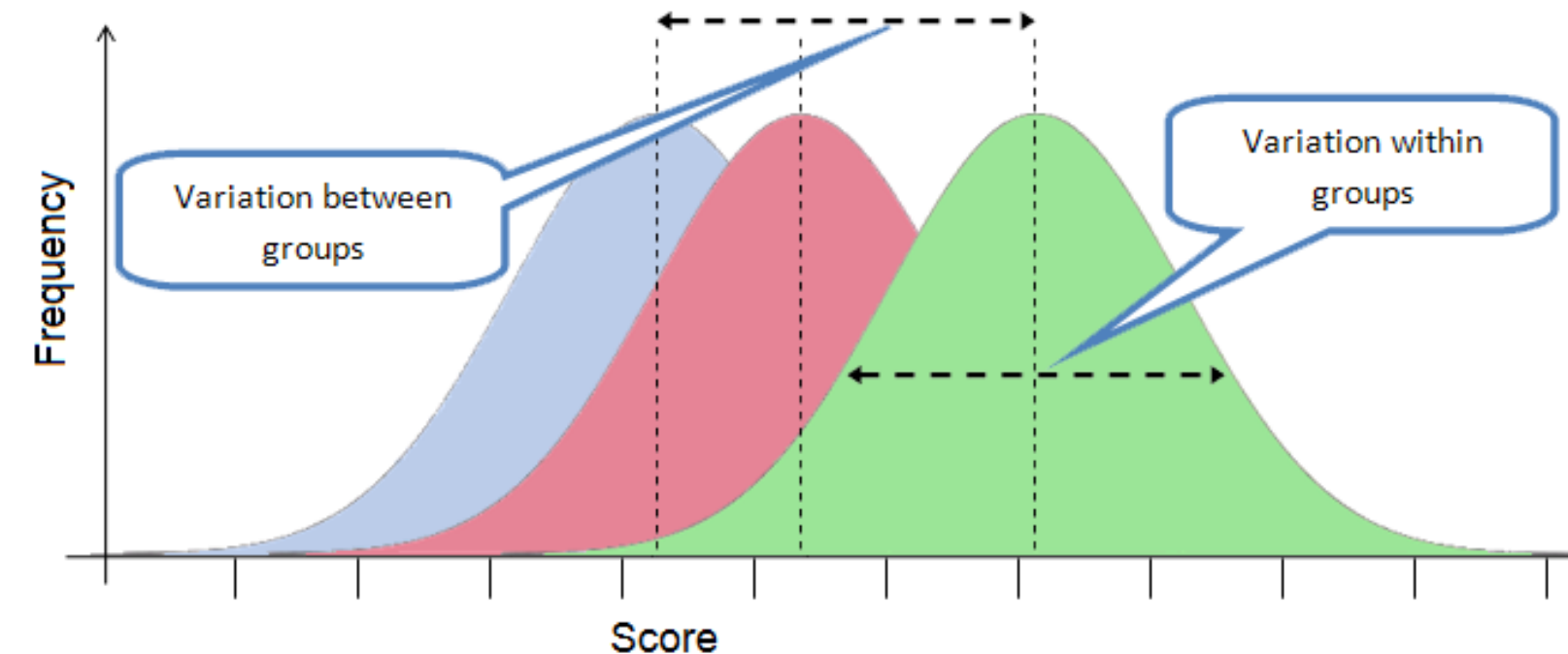
How spread out is data
from the overall
average?

Between-Group Variation

- It calculates the **distance between each group's individual mean** and the **overall mean** of the entire dataset (the "Grand Mean").
- The between-group variation is denoted by :

$$SS_{between} = \sum n_i (\bar{x}_i - \bar{x})^2$$

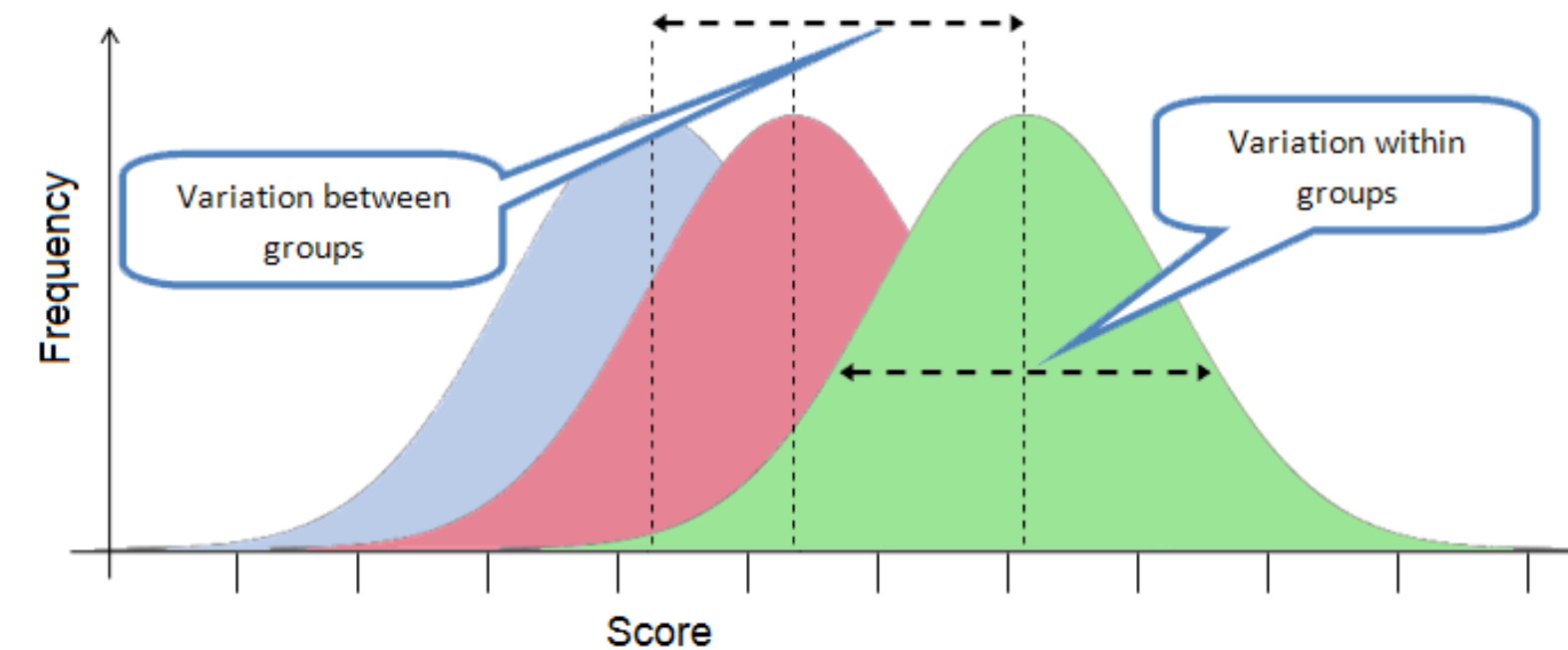
- n_i = size of group i
- $\bar{x}_i$ = mean of group i
- $\bar{x}$ = overall mean



Within-Group Variation

- It tells us how much values vary within the same group.
- It is represented as

$$SS_{within} = \sum \sum (x_{ij} - \bar{x}_i)^2$$



The ANOVA F-Statistic Formula

$$F = \frac{\text{Variance Between Groups}}{\text{Variance Within Groups}}$$

$$F = \frac{MS_{between}}{MS_{within}}$$

- Higher F ratio values indicate the variation between groups is larger than the individual variation of groups.
- In such cases, it is more likely that the mean of the groups are different.

The ANOVA F-Statistic Formula

$$MS_{between} = \frac{SS_{between}}{df_{between}}$$

Degrees of freedom:

$$df_{between} = k - 1$$

(k = number of groups)

The ANOVA F-Statistic Formula

$$MS_{within} = \frac{SS_{within}}{df_{within}}$$

Degrees of freedom:

$$df_{within} = N - k$$

(N = total observations)

Why ANNOVA?

A time series can be seen as:

$$y_t = (\text{level}) + (\text{seasonality}) + (\text{trend}) + (\text{noise})$$

ANOVA helps answer:

Is there evidence that the level differs across groups like:

- Month (Jan vs Feb vs ...)
- Day-of-week (Mon vs Tue vs ...)
- Pre vs Post intervention
- Festival vs non-festival periods
- Before policy change vs after
- So it becomes a mean comparison tool for time-structured categories.

Annova is used

- To detect seasonality
- To test impact of an intervention / change point (A/B in time)
- To compare multiple time-based regimes

How to Use ANNOVA

How to use ANOVA correctly in time-series workflows

ANOVA on residuals

Common approach:

1. Fit time-series structure first (trend/ARIMA)
2. Take residuals (should be uncorrelated)
3. **Apply ANOVA on residuals grouped by season/month/etc.**

ANOVA is useful in time series when you want to test whether the mean level changes across time-based groups or regimes (seasonality, interventions), but you must account for autocorrelation to avoid misleading significance.

Error

- Predictions without error measurement are meaningless.
- It is essential to recognize our mistakes.
- Always when:
 - You make predictions
 - Decisions depend on correctness.

Error

Situation	Risk
Only average error is low	Rare disasters ignored
Only percentage error used	Breaks when values ≈ 0
Symmetric error assumed	Underprediction may be costlier

Mean Absolute Error (MAE)

- It tells us, on average, **how far off** we are.
- **Interpretation**
 - **Same unit** as data
 - Easy to **explain to non-technical stakeholders**
- **Strengths**
 - Robust to **outliers**
 - **Intuitive**
- **Limitations**
 - Treats all **errors equally**
 - Does not emphasize **big mistakes**

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

Mean Squared Error (MSE)

- It tells us how costly our mistakes are if big errors matter more.
- **Interpretation**
 - **Squaring** increases the **penalty** for **large errors**
 - Used heavily in **optimization**
- **Strengths**
 - Strong penalty for big errors
- **Limitations**
 - Hard to interpret (squared units)
 - Hard to interpret (squared units)

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

Root Mean Squared Error (RMSE)

- It tells us how large errors are when big mistakes matter
- **Interpretation**
 - Same **unit** as data
 - Highlights volatility in prediction quality
- **Strengths**
 - Penalizes big errors
 - Interpretable scale
- **Limitations**
 - sensitive to outliers

$$RMSE = \sqrt{MSE}$$

Problem

- Suppose we are **forecasting** daily **electricity demand** (in units) for 5 days.
- Question – Compute the **Mean Absolute error** for this data.

Day	Actual Demand	Predicted Demand
1	100	98
2	120	125
3	130	128
4	150	140
5	170	180

Solution

Day	Actual Value	Predicted Value	Error = Actual - Predicted
1	100	98	2
2	120	125	-5
3	130	128	2
4	150	140	10
5	170	180	-10

$$MAE = \frac{|2| + |-5| + |2| + |10| + |-10|}{5} = 5.8$$

Comparing 2 Models

- Suppose we have actual sales for 5 days and predictions from two models:

- Model A
- Model B

Which Model is better?

Day	Actual Sales	Model A Prediction	Model B Prediction
1	100	95	100
2	120	118	110
3	130	125	140
4	150	145	120
5	170	165	200

Comparing 2 Models

$$MAE_A = \frac{5 + 2 + 5 + 5 + 5}{5} = 4.4$$

$$MAE_B = \frac{0 + 10 + 10 + 30 + 30}{5} = 16$$

- Model A is much more reliable on average.

Day	Actual Sales	Model A Prediction	Model B Prediction
1	100	95	100
2	120	118	110
3	130	125	140
4	150	145	120
5	170	165	200

Comparing 2 Models

$$RMSE_A = \sqrt{\frac{25 + 4 + 25 + 25 + 25}{5}} = 4.456$$

$$RMSE_B = \sqrt{\frac{0 + 100 + 100 + 900 + 900}{5}} = 20$$

- Model B has dangerously large errors, even if sometimes perfect.

Day	Actual Sales	Model A Prediction	Model B Prediction
1	100	95	100
2	120	118	110
3	130	125	140
4	150	145	120
5	170	165	200

Comparing 2 Models

- Model A behaves predictably.
- Model B behaves erratically.
- Model A will be selected because
 - Lower average error
 - No catastrophic failures
 - Easier to plan inventory and staffing

Day	Actual Sales	Model A Prediction	Model B Prediction
1	100	95	100
2	120	118	110
3	130	125	140
4	150	145	120
5	170	165	200

Seasonal ARIMA (SARIMA)

- Regular ARIMA learns:
 - **Trend**
 - Short-term dependence
- But it **cannot remember** repeating **cycles** like:
 - Weekly routines
 - Monthly patterns
 - Yearly seasons
- **Seasonal ARIMA** adds **memory** of the same time in **previous cycles**.

SARIMAX Model

SARIMAX = SARIMA + eXogenous variables

- Used when data has trend, seasonality, and external factors
- **It models:**
 - Past values (AR)
 - Past errors (MA)
 - Differencing (I)
 - Seasonal patterns (S)
 - External variables (X)
- **Best suited when:**
 - Data shows seasonality (daily, monthly, yearly)
 - External factors influence the series
 - Simple ARIMA is not sufficient

Other time series models

ARIMAX Model

- Used when the time series is influenced by external factors
- **Uses:**
 - Past values of the series
 - Past errors
 - Other related variables (X)
- Best suited when:
 - External variables affect the data
 - Better accuracy than ARIMA is needed
 - Future X values are available

VAR Model (Vector Autoregression)

- **VAR models multiple time series together**
 - Each variable is a function of its own past values and the past values of all other variables
 - All variables are treated as endogenous (no strict input/output)
- **Best suited when:**
 - Several time series influence each other
 - Variables move together over time
 - System-level forecasting is required

From ARIMA to LSTM

- In time series, we always try to use the past to predict the future
- AR, MA, ARIMA do this using a fixed number of past days and errors

But in real data, we often don't know:

- how far back to look
- which past patterns really matter

LSTM is a deep learning model that learns this automatically

- It keeps a memory of the past and decides:
 - what to remember
 - what to forget
 - what is important for prediction

Problem to solve

We are given the following time series data for variable X_t and the corresponding white noise/error terms ϵ_t :

Time t	X_t	ϵ_t (White Noise/Error Term)
1	5	0.5
2	6	1.0
3	7	-0.3
4	?	0.4

We need to predict X_4 using the following models:

1. Autoregression (AR(1)) model:

The AR(1) model is given by:

$$X_t = c + \phi_1 X_{t-1} + \epsilon_t$$

Where:

- c is the intercept term,
- ϕ_1 is the autoregressive parameter,
- ϵ_t is the white noise error term for time t .

2. Moving Average (MA(1)) model:

The MA(1) model is given by:

$$X_t = c + \theta_1 \epsilon_{t-1} + \epsilon_t$$

Where:

- c is the intercept term,
- θ_1 is the moving average parameter,
- ϵ_{t-1} is the previous error term.

3. ARIMA(1,1,1) model:

The ARIMA(1,1,1) model incorporates autoregression, differencing, and moving average. For this example, we assume $\phi_1 = 0$. The model is given by:

$$\Delta X_t = c + \theta_1 \epsilon_{t-1} + \epsilon_t$$

Where:

$$\Delta X_t = X_t - X_{t-1} \quad \text{is the differenced series.}$$

Task:

- Calculate the intercept term c along with the coefficients ϕ_1 and θ_1 for each model.
- Using the models, predict the value of X_4 .

Problem

Consider the following bivariate time series data:

t	X_t	Y_t
1	2	1
2	4	2
3	5	3
4	8	6
5	X_5	Y_5

Assume that the variable X_t follows a **first-order Vector Autoregressive structure (VAR(1))** with a non-zero intercept and zero error terms.

Model Specification

The model for X_t is given by:

$$X_t = C + \phi_1 X_{t-1} + \phi_2 Y_{t-1}, \quad t \geq 2,$$

where:

- C is the intercept term,
- ϕ_1 and ϕ_2 are autoregressive coefficients,
- the error term is assumed to be zero for simplicity.

Task

1. Use the given observations to estimate the parameters C , ϕ_1 , and ϕ_2 .
2. Using the estimated model, compute the value of X_5 .

Case Study

Urban Air Quality & Public Health Forecasting

US Government Time Series Case Study

Your Role: Government Data Scientist

You have just joined a **US Government Public Health & Climate Analytics Unit**.

Over the past few years, hospitals across major US cities have reported a steady rise in:

- Heat stroke cases
- Breathing disorders
- Asthma attacks
- Cardiac emergencies

Government officials suspect that **weather conditions and air quality are major hidden drivers** behind these public health problems.

To investigate this, the government has provided you with a nationwide dataset called:

Urban Air Quality and Health Impact Dataset

It contains daily environmental and weather records along with a computed **Health_Risk_Score**.

Please fill the feedback form.

Thank You